

Integration — Lesson 39 Test: Parametric & Polar

10 questions · show your work in the box · angles in radians · exact answers with π and $\sqrt{}$ are fine

Name: _____ Date: _____ Score: _____ / 10

1. A point moves with $x = 2t$, $y = t - 3$ for t in $[0, 4]$. (a) Give the point at $t = 0$, $t = 2$, ^{1 pt} and $t = 4$. (b) Eliminate the parameter to find an equation in x and y , and name the curve.

2. A point moves with $x = t^2$, $y = 8t$. (a) Find dx/dt and dy/dt . (b) Find the speed at $t = 3$. ^{1 pt}

3. A point moves with $x = 5t + 2$, $y = 12t - 1$. (a) Show that its speed is the same at ^{1 pt} every time, and find that speed. (b) Find the distance it travels from $t = 1$ to $t = 4$.

4. A point moves with $x = 3\cos t$, $y = 3\sin t$ for t in $[0, 2\pi]$. (a) Find its speed. (b) 1 pt
Use $L = \int \text{speed } dt$ to find the total distance travelled. (c) Confirm your answer using the circumference of a circle.

5. A point moves with $x = 4\cos 2t$, $y = 4\sin 2t$ for t in $[0, 2\pi]$. (a) Find its speed. (b) 1 pt
Find the total distance travelled. (c) The circle it moves on has circumference 8π — explain why your answer to (b) is larger. (d) At what time does the point finish its first lap?

6. A point moves with $x = 3t^2$, $y = 2t^3$ for t in $[0, \sqrt{3}]$. Find the exact distance it travels. 1 pt
(Factor the speed first, then use a u-substitution.)

7. (a) Describe the graph of $r = 4$ in words. (b) Describe the graph of $\theta = \pi/6$ in words. 1 pt
(c) Give the ordinary (x, y) coordinates of the polar point $(r, \theta) = (4, \pi/6)$. (d) Make a table of $r = 2 + 2\cos \theta$ at $\theta = 0, \pi/2, \pi, 3\pi/2$ and name the curve.

8. (a) Use $A = \frac{1}{2} \int r^2 d\theta$ to find the area inside $r = 6$, sweeping θ from 0 to 2π , and confirm it with the ordinary circle-area formula. (b) Now find the area swept as θ goes only from 0 to $\pi/2$, and check that it is the fraction of the circle you expect. 1 pt

9. Find the exact area inside the circle $r = 2\cos \theta$. Sweep θ from $-\pi/2$ to $\pi/2$, and explain in one sentence why those are the correct limits. Show every step of the half-angle identity you use, and check your answer against the geometry of the circle. 1 pt

10. A student makes two polar-area attempts. (a) For $r = 5$ the student writes " $A = \int_0^{2\pi} 5 d\theta = 10\pi$." Identify the error, say what $\int r d\theta$ actually measures, and give the correct area. (b) The same student computes the area inside $r = 3\cos \theta$ as $\frac{1}{2} \int_0^{2\pi} (3\cos \theta)^2 d\theta$ and gets $9\pi/2$, but the true area is $9\pi/4$. Identify this second error and give the correct computation. 1 pt
